

THE STRUCTURE OF MATHEMATICAL REALITY

Phase III — Chapter 8

Vector Spaces and the Meaning of “Canvas”

Parallel Reading

Self-Study Curriculum

Parallel Reading: Sheldon Axler, *Linear Algebra Done Right* (3rd or 4th ed.), Chapters 1-3 (approximately 90 pages). This book teaches linear algebra starting from vector spaces rather than matrices, which is exactly the perspective needed here.

Also read: PCM sections III.50 (“Linear Operators and Their Properties”) and III.37 (“Hilbert Spaces”): brief reference entries; the second connects inner products to the quantum-mechanical state spaces this lecture mentions.